Remote and Distributed Quantum Hypothesis Testing

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***Abstract*—We characterize the Stein exponent of a remote quantum hypothesis testing problem where, depending on the hypothesis, a remote sensor observes one of two quantum states and communicates to the decision center over a quantum channel. We show a dichotomy of the Stein exponent depending on whether the orthogonal Kraus space of the channel contains a rank-one matrix or not. In the former case and when the admissible type-I error probability is sufficiently large, the same Stein exponent is achievable as over a perfect channel whereas in the latter case the Stein exponent is zero for one channel use. Interesting phenomena occur when the channel can be used multiple times. For channels with orthogonal Kraus spaces containing rank-one matrices, the perfect-channel Stein exponent becomes achievable at smaller type-I error thresholds, and the threshold is even allowed to vanish as the number of channel uses increases. For channels with orthogonal Kraus spaces not containing rank-one matrices, self-superactivation can occur, i.e., for some channels the orthogonal Kraus space of a single channel use contains no rank-one matrices, but for two channel uses it does, allowing to achieve the perfect-channel Stein exponent. We show that self-superactivation can occur for a channel with input/output dimension 4 but not for channels where the orthogonal Kraus space has dimension at most one, and in particular not for qubit channels or interior channels. For the latter, the Stein exponent remains equal to zero, even when the number of channel uses grows sublinearly with the number of observations.**

I. Introduction

Consider a quantum binary hypothesis testing problem. Under hypothesis $\mathcal{H}=0$, one observes n copies of a quantum state ρ, whereas under $\mathcal{H}=1$, one observes n copies of another state σ. The goal is to decide which hypothesis is true, with small probabilities of type-I error (guessing $\hat{\mathcal{H}}=1$ when $\mathcal{H}=0$) and type-II error (guessing $\hat{\mathcal{H}}=0$ when $\mathcal{H}=1$). Our main focus is on the *Stein exponent*, which measures the asymptotic exponential decay rate (as $n\to\infty$) of the type-II error probability under a given threshold ϵ on the allowed type-I error. When the quantum states are directly available at the decision center, the optimal Stein exponent equals the quantum relative entropy $D(\rho\|\sigma)$ [7], [9].

In this work, we consider a remote setting where a sensor observes the n quantum systems, and communicates to the decision center over k uses of a quantum channel $\mathcal{N}$, see Figure 1. We study both a fixed number of channel uses and a number $k=k(n)$ that grows to infinity but sublinearly in n. The latter assumption models scenarios where the data is high-dimensional but communication resources are scarce. A slightly more general setup was considered in [3], [13] in the sense that the decision center also had access to one part of the n copies of the bipartite quantum states ρ or σ. Most related to our work is [13] which considered a setup with a remote sensor that communicates to the decision center over a sublinear number of perfect channel uses. Under this assumption, the Stein exponent was derived for several classes of states ρ and σ.

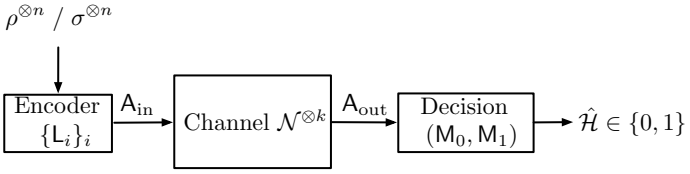


Fig. 1: Remote quantum hypothesis testing.

More relevant literature can be found in the classical case [1], [2], [6], [8], [12]. In particular, [2] considers shared observations between a remote sensor and the decision center and shows a dichotomy in the Stein exponent when communication is over a Discrete Memoryless Channel (DMC) that can be used a sublinear number of times. If the DMC is fully-connected, the Stein exponent collapses to the local exponent at the decision center, and when it is partially-connected it is equal to the exponent over the perfect channel.

In this work, we show that a similar dichotomy holds in the quantum case when only the remote sensor observes n copies of a quantum state. Specifically, assume that the employed quantum channel $\mathcal{N}$ can be used k times and has effective output space equal to true output space. We show that if the orthogonal Kraus space $\mathcal{K}^{\perp}_{\mathcal{N}^{\otimes k}}$ of the k-fold product channel $\mathcal{N}^{\otimes k}$ contains a rank-one matrix, then the perfect-channel Stein exponent is achievable for sufficiently large type-I error thresholds; otherwise the Stein exponent is zero. In particular, quantum channels in the relative interior of the set $\mathcal{S}$ of Completely Positive Trace-Preserving (CPTP) maps with full effective output space have a zero Stein exponent, even if the number of channel uses grows sublinearly with n. On the boundary of $\mathcal{S}$, an interesting self-superactivation phenomenon can occur and the orthogonal Kraus space $\mathcal{K}^{\perp}_{\mathcal{N}}$ of some channels $\mathcal{N}$ does not contain a rank-one matrix but $\mathcal{K}^{\perp}_{\mathcal{N}^{\otimes 2}}$ of the two-fold product channel does. We give an explicit example with

four-dimensional input and output spaces. We also establish cases in which this self-superactivation is impossible, including qubit channels and channels with a one-dimensional orthogonal Kraus space.

II. Problem Setup

Consider the remote hypothesis testing problem in Figure 1. For a given blocklength n, a sensor observes the quantum state $\rho^{\otimes n}$ under $\mathcal{H}=0$ and it observes the quantum state $\sigma^{\otimes n}$ under $\mathcal{H}=1$, where we assume $D(\rho\|\sigma) := \mathrm{Tr}\left(\rho(\log\rho-\log\sigma)\right) < \infty$. The channel $\mathcal{N}$ is of input dimension N_{in} and output dimension N_{out} and it is represented by the Kraus operators $\{\mathsf{K}_j\}_j$. It can be used k times, where depending on the scenario k can be a given positive integer, or also a sequence $k=k(n)$ that grows with n.

For given k, the sensor applies a CPTP operation represented by Kraus operators $\{\mathsf{L}_i\}_i$ to the observed quantum state (either $\rho^{\otimes n}$ or $\sigma^{\otimes n}$) to produce the input state $\mathsf{A}_{\text{in}} \in \mathbb{C}^{N_{\text{in}}^k \times N_{\text{in}}^k}$. Under $\mathcal{H}=0$ the channel input state is thus

$$\mathsf{A}_{\text{in}}^{\rho} := \sum_i \mathsf{L}_i \rho^{\otimes n} \mathsf{L}_i^{\dagger} \tag{1a}$$

and under $\mathcal{H}=1$ it is

$$\mathsf{A}_{\text{in}}^{\sigma} := \sum_i \mathsf{L}_i \sigma^{\otimes n} \mathsf{L}_i^{\dagger}. \tag{1b}$$

The decoder observes the output state of the k uses of the channel, $\mathsf{A}_{\text{out}} \in \mathbb{C}^{N_{\text{out}}^k \times N_{\text{out}}^k}$ and applies a binary measurement $(\mathsf{M}_0, \mathsf{M}_1)$ of its choice to produce the binary output $\hat{\mathcal{H}} \in \{0,1\}$.

A type-I error occurs when the decision center decides $\hat{\mathcal{H}}=1$ under $\mathcal{H}=0$, whereas a type-II error occurs when it decides $\hat{\mathcal{H}}=0$ under $\mathcal{H}=1$. The probabilities of these error events are denoted α_n and β_n, respectively. We thus have:

$$\alpha_n \triangleq 1 - \mathrm{Tr}\left(\mathsf{M}_0 \mathcal{N}^{\otimes k}\left(\mathsf{A}_{\text{in}}^{\rho}\right)\right) \tag{2}$$

and

$$\beta_n \triangleq \mathrm{Tr}\left(\mathsf{M}_0 \mathcal{N}^{\otimes k}\left(\mathsf{A}_{\text{in}}^{\sigma}\right)\right). \tag{3}$$

Definition 1 (Fixed k-achievability)**.** *Given a positive integer k, the* Stein exponent $\theta_k^{\star}(\epsilon)$ *is defined as the supremum over all values $\theta \geq 0$ for which there exists a sequence (in n) of encodings $\{\mathsf{L}_i\}_i$ for the k-fold channel $\mathcal{N}^{\otimes k}$ and corresponding output measurements $(\mathsf{M}_0, \mathsf{M}_1)$ so that*

$$\overline{\lim_{n\to\infty}}\ \alpha_n \leq \epsilon \tag{4a}$$

$$-\overline{\lim_{n\to\infty}}\ \frac{1}{n}\log\beta_n \geq \theta. \tag{4b}$$

Definition 2 (Achievability with Sublinear k)**.** *Let now $k(n)$ be a sequence satisfying*

$$\lim_{n\to\infty} k(n) = \infty \tag{5a}$$

$$\lim_{n\to\infty} \frac{k(n)}{n} = 0. \tag{5b}$$

Then, the Stein exponent $\theta_{\text{sublin}}^{*}(\epsilon)$ is defined as the supremum over all values $\theta \geq 0$ for which it is possible to find for each n an encoding $\{\mathsf{L}_i\}_i$ for the $k(n)$-fold channel $\mathcal{N}^{\otimes k(n)}$ and corresponding output measurements $(\mathsf{M}_0, \mathsf{M}_1)$ satisfying (4)*.*

III. Results

The following technical lemma will be used in the proofs of our converse results.

Lemma 1. *For fixed blocklength n and number of channel uses k, the type-II error probability β_n can be bounded in terms of the type-I error probability α_n as:*

$$\beta_n \geq \gamma_{\mathcal{N},k}(1-\alpha_n), \tag{6}$$

where

$$\gamma_{\mathcal{N},k} := \min_{\substack{\tau\succeq 0:\\ \mathrm{Tr}(\tau)=1}} c_{\min}\left(\mathcal{N}^{\otimes k}(\tau)\right), \tag{7}$$

for $c_{\min}(\mathsf{A})$ the smallest eigenvalue of a matrix A.

Proof. See Section IV-A. □

To present our results, we partition the set of CPTP maps $\mathcal{Q}$ into different subsets. Our first result concerns the relative interior $\mathcal{Q}^{\circ}$ of $\mathcal{Q}$, when the latter is considered a subset of all trace- and Hermiticity-preserving maps.

A. Channels in the Relative Interior $\mathcal{Q}^{\circ}$

Start by noticing that any quantum channel $\mathcal{N} \in \mathcal{Q}^{\circ}$ has a full-rank Choi matrix $\mathsf{J}_{\mathcal{N}}$ and a minimal Kraus representation of $\mathcal{N}$ thus consists of $N_{\text{in}}N_{\text{out}}$ Kraus elements $\mathsf{K}_1, \ldots, \mathsf{K}_{N_{\text{in}}N_{\text{out}}}$. The channel's Kraus span

$$\mathcal{K}_{\mathcal{N}} := \mathrm{Span}\{\mathsf{K}_i\}_i \tag{8}$$

thus equals the entire matrix space $\mathbb{C}^{N_{\text{out}}\times N_{\text{in}}}$.

Our first result is a complete characterization of the Stein exponent for interior channels: it equals zero, irrespective of ϵ and k.

Theorem 1 (Zero Stein Exponent for Interior Channels)**.** *If $\mathcal{N} \in \mathcal{Q}^{\circ}$ then the Stein exponent vanishes for any k:*

$$\theta_k^{*}(\epsilon) = 0, \quad \forall k \geq 1,\ \epsilon \in [0,1). \tag{9}$$

Moreover, the sublinear Stein exponent also vanishes

$$\theta_{\text{sublin}}^{*}(\epsilon) = 0, \quad \forall \epsilon \in [0,1). \tag{10}$$

Proof. Follows by Lemma 1 and by showing that for interior channels $\gamma_{\mathcal{N},k} \geq a^k$ for an appropriate $a > 0$. Details of the proof are given in Section IV-B. □

B. Channels on the Boundary of $\mathcal{S}$ and in $\mathcal{C}$

The rest of the result section focuses on the relative boundary of $\mathcal{Q}$. We wish to limit attention to a subset of this boundary to avoid degenerate situations. For this purpose, introduce the following subclass of CPTP maps:

$$\mathcal{S} := \left\{ \mathcal{N} \in \mathcal{Q} \;\middle|\; \mathrm{rank}(\mathcal{N}(\mathsf{I})) = N_{\text{out}} \right\}. \tag{11}$$

(The subset $\mathcal{S}$ also contains channels in the relative interior of $\mathcal{Q}$. Since we already determined all relevant Stein exponents in the relative interior, our focus will however be on the boundary channels.)

Notice that the condition $\operatorname{rank}(\mathcal{N}(\mathsf{I})) = N_{\text{out}}$ is no stronger than requiring that $\mathcal{N}(\kappa)$ is full-rank for *some* input state κ. Thus, $\mathcal{S}$ simply restricts to channels that leave no output direction unexploited.

Remark 1. *For any channel $\mathcal{N} \in \mathcal{S}$ and any matrix $\mathsf{M} \in \mathbb{C}^{N_{\text{out}} \times N_{\text{out}}}$ with $\mathsf{I} \succeq \mathsf{M} \succeq 0$ and $\mathsf{M} \neq 0$, there exists a rank-one input $|\phi\rangle\langle\phi|$ such that*

$$\operatorname{Tr}\left(\mathsf{M}\mathcal{N}(|\phi\rangle\langle\phi|)\right) > 0. \tag{12}$$

Proof. Since $\mathcal{N} \in \mathcal{S}$, the matrix $\mathcal{N}(\mathsf{I})$ is positive definite. Thus $\operatorname{Tr}\left(\mathsf{M}\mathcal{N}(\mathsf{I})\right) > 0$ for any nonzero positive semidefinite matrix M. Expanding I in an orthonormal basis $\{|j\rangle\}$ gives

$$\sum_j \operatorname{Tr}\left(\mathsf{M}\mathcal{N}(|j\rangle\langle j|)\right) > 0. \tag{13}$$

Each summand is nonnegative, so at least one is positive. □

We partition the set $\mathcal{S}$ into the subset $\mathcal{C}$ and its complement $\bar{\mathcal{C}}$, depending on whether the orthogonal Kraus space (with respect to the Hilbert–Schmidt inner product)

$$\mathcal{K}_{\mathcal{N}}^{\perp} := \{\tau \colon \operatorname{Tr}\left(\tau^{\dagger}\mathsf{K}\right) = 0, \quad \forall \mathsf{K} \in \mathcal{K}_{\mathcal{N}}\} \tag{14}$$

contains a rank-one matrix or not. Define thus:

$$\mathcal{C} := \left\{\mathcal{N} \in \mathcal{S} \mid \exists \tau \in \mathcal{K}_{\mathcal{N}}^{\perp},\ \operatorname{rank}(\tau) = 1\right\}, \tag{15}$$

$$\bar{\mathcal{C}} := \mathcal{S} \setminus \mathcal{C}. \tag{16}$$

Remark 2 (Discussion on the Classes of Channels)**.** *All channels on the relative interior of $\mathcal{S}$ (with respect to the set of trace- and Hermiticity-preserving maps) belong to the set $\bar{\mathcal{C}}$ (and also to the relative interior of $\mathcal{Q}$). In fact, interior channels $\mathcal{N} \colon \mathbb{C}^{N_{\text{in}} \times N_{\text{in}}} \to \mathbb{C}^{N_{\text{out}} \times N_{\text{out}}}$ have full Kraus space $\mathcal{K} = \mathbb{C}^{N_{\text{out}} \times N_{\text{in}}}$ and thus the orthogonal Kraus space $\mathcal{K}^{\perp} = \{0\}$ is degenerate and does not contain a nonzero rank-one matrix. Channels on the relative boundary of $\mathcal{S}$ belong either to $\mathcal{C}$ or to $\bar{\mathcal{C}}$. For example, the perfect channel $\mathcal{N}_1 \colon \tau \mapsto \tau$ has Kraus space $\mathcal{K}_{\mathcal{N}_1} = \operatorname{Span}\{\mathsf{I}\}$ and thus the orthogonal complement $\mathcal{K}_{\mathcal{N}_1}^{\perp}$ contains all rank-one matrices E_{ij} for $i \neq j$ and E_{ij} the matrix with all-zeros except entry one in row i and column j. Channel $\mathcal{N}_1$ thus belongs to $\mathcal{C}$. The amplitude damping channel is another example of a channel that belongs to $\mathcal{C}$. On the other hand, the qubit channel*

$$\mathcal{N}(\tau) = 1/3\tau + 1/3\mathsf{X}\tau\mathsf{X} + 1/3\mathsf{Z}\tau\mathsf{Z}, \tag{17}$$

where X and Z are the Pauli matrices, lies on the boundary of $\mathcal{S}$ and belongs to $\bar{\mathcal{C}}$. In fact, the Kraus operators of this channel are $\mathsf{K}_0 = \sqrt{1/3}\mathsf{I}_2$, $\mathsf{K}_1 = \sqrt{1/3}\mathsf{X}$, and $\mathsf{K}_2 = \sqrt{1/3}\mathsf{Z}$ and its orthogonal Kraus space is thus $\mathcal{K}_{\mathcal{N}}^{\perp} = \operatorname{Span}\{\mathsf{Y}\}$, which does not contain a nonzero rank-one matrix. A second example is the channel $\mathcal{N}_W$ defined through its Kraus operators in (32) *ahead, see also Theorem 7.*

Lemma 2. *For any quantum channel $\mathcal{N} \in \mathcal{C}$ there exists a rank-one input state $|\psi\rangle\langle\psi|$ such that $\operatorname{Tr}\left(\mathsf{M}\mathcal{N}(|\psi\rangle\langle\psi|)\right) = 0$ for some rank-one matrix $\mathsf{M} \succeq 0$ with $\mathsf{M} \neq 0$.*

Proof. Let $\mathsf{M} = |v\rangle\langle v|$ and notice the equivalences

$$\begin{aligned}
& \operatorname{Tr}\left(\mathsf{M}\mathcal{N}(|\psi\rangle\langle\psi|)\right) = 0 \\
\Leftrightarrow \quad & \operatorname{Tr}\left(|v\rangle\langle v| \sum_i \mathsf{K}_i |\psi\rangle\langle\psi| \mathsf{K}_i^{\dagger}\right) = 0 && (18)\\
\Leftrightarrow \quad & \sum_i |\langle v|\mathsf{K}_i|\psi\rangle|^2 = 0 && (19)\\
\Leftrightarrow \quad & \forall i\colon\ \langle v|\mathsf{K}_i|\psi\rangle = 0 && (20)\\
\Leftrightarrow \quad & \forall i\colon \operatorname{Tr}\left(|\psi\rangle\langle v|\mathsf{K}_i\right) = 0 && (21)\\
\Leftrightarrow \quad & |v\rangle\langle\psi| \in \mathcal{K}^{\perp}. && (22)
\end{aligned}$$

□

Lemma 2 and Remark 1 imply that for each $\mathcal{N} \in \mathcal{C}$ it is possible to find $|\phi\rangle$ and $|\psi\rangle$ and a matrix $\mathsf{I} \succeq \mathsf{M} \succeq 0$ such that

$$\operatorname{Tr}\left(\mathsf{M}\mathcal{N}(|\psi\rangle\langle\psi|)\right) = 0 \tag{23a}$$

$$\operatorname{Tr}\left(\mathsf{M}\mathcal{N}(|\phi\rangle\langle\phi|)\right) > 0. \tag{23b}$$

This is key to prove the achievability results in the following theorem. The converse follows trivially, since the Stein exponent cannot exceed the local exponent:

$$\theta_k^*(\epsilon), \theta_{\text{sublin}}^*(\epsilon) \leq D(\rho\|\sigma). \tag{24}$$

Theorem 2. *If $\mathcal{N} \in \mathcal{C}$ then $\forall k \in \mathbb{N}^+,\ \exists q \in [0,1)$ such that:*

$$\theta_k^*(\epsilon) = D(\rho\|\sigma), \quad \forall \epsilon \in [q^k, 1). \tag{25}$$

Moreover, with a sublinear number of channel uses[1]

$$\theta_{\text{sublin}}^*(\epsilon) = D(\rho\|\sigma), \quad \forall \epsilon \in [0,1). \tag{26}$$

Proof. Pick pure states $\psi = |\psi\rangle\langle\psi|$ and $\phi = |\phi\rangle\langle\phi|$ and a matrix M satisfying (23), as well as a sequence of local measurements at the sensor $\{(\Pi_{0,n}, \Pi_{1,n})\}$ that achieves the local quantum Stein exponent, i.e., so that applying the measurement to $\rho^{\otimes n}$ or $\sigma^{\otimes n}$ results in $\alpha_{n,\text{local}} \to 0$ as $n \to \infty$ and $-\overline{\lim}\,\frac{1}{n}\log\beta_{n,\text{local}} = D(\rho\|\sigma)$. Apply this binary measurement to the sensor's observation $\rho^{\otimes n}$ or $\sigma^{\otimes n}$ and for outcome $B = 0$ send $\mathsf{A}_{\text{in}} = \phi^{\otimes k}$ while for outcome $B = 1$ send $\mathsf{A}_{\text{in}} = \psi^{\otimes k}$. The decision center applies the binary measurement $(\mathsf{I}-(\mathsf{I}-\mathsf{M})^{\otimes k}, (\mathsf{I}-\mathsf{M})^{\otimes k})$ to the channel output A_{out} and declares the outcome of this measurement as the hypothesis $\hat{\mathcal{H}}$.

The described scheme is analyzed in Section IV-C. It is shown (using (23)) that the type-II error probability β_n of the overall scheme cannot exceed the type-II error probability $\beta_{n,\text{local}}$ of the local test at the sensor, while the type-I error probability α_n incurs a penalty compared to $\alpha_{n,\text{local}}$ that however can vanish with increasing k. □

In the literature, matrix spaces without rank-one elements are also known as *completely entangled* [10] and matrix spaces whose complements do not contain a rank-one matrix as *transitive spaces* [5]. The following corollary directly follows

[1]Even though beyond the scope of this manuscript, the result in (26) remains valid when $\overline{\lim}_{n\to\infty} k(n)/n > 0$, simply because no test can outperform the local test and thus improve over the exponent $D(\rho\|\sigma)$.

from the result in [5, Corollary 3.4] on transitive matrix spaces and from Theorem 2.

Corollary 3. *Any quantum channel $\mathcal{N} \in \mathcal{S}$ with*

$$\dim(\mathcal{K}_{\mathcal{N}}) < N_{\text{in}} + N_{\text{out}} - 1 \tag{27}$$

belongs to $\mathcal{C}$ and thus for this channel $\exists q \in [0,1)$ such that:

$$\theta_k^*(\epsilon) = D(\rho\|\sigma), \quad \forall \epsilon \in [q^k, 1). \tag{28}$$

We conclude this subsection by remarking that all classical channels in $\mathcal{S}$ belong to either $\mathcal{C}$ or to $\mathcal{S}^\circ$ because they can be described by the rank-one Kraus operators $\mathsf{K}_{y,x} = \sqrt{P_{y|x}}\,|y\rangle\langle x|,\ \forall x, y$. When the classical channel is fully-connected, i.e., $P_{y|x} > 0$ for all (x, y), then the channel belongs to $\mathcal{S}^\circ$. If it is partially connected, i.e., $P_{y|x} = 0$ for some (x, y), then the orthogonal Kraus space $\mathcal{K}^\perp$ contains the rank-one element $|y\rangle\langle x|$ and thus the channel belongs to $\mathcal{C}$. Theorems 1 and 2 fully characterize all Stein exponents $\theta_k^\star(\epsilon)$ and $\theta_{\text{sublin}}^\star(\epsilon)$ for all classical channels in $\mathcal{S}$ and the next subsection concerns only purely quantum channels.

C. Channels on the Boundary of $\mathcal{S}$ and in $\bar{\mathcal{C}}$

The following result combined with Theorem 2 provides a complete characterization of the Stein exponent for $k = 1$ for all channels $\mathcal{N} \in \mathcal{S}$.

Theorem 4. *If $\mathcal{N} \in \bar{\mathcal{C}}$ then*

$$\theta_1^*(\epsilon) = 0 \quad \forall \epsilon \in [0, 1). \tag{29}$$

Proof. See Section IV-D. □

We observe a dichotomy: for $\mathcal{N} \in \mathcal{C}$ the Stein exponent $\theta_1^\star(\epsilon)$ coincides with $D(\rho\|\sigma)$ achieved over a perfect quantum channel whenever ϵ exceeds a channel-dependent threshold, while for $\mathcal{N} \in \bar{\mathcal{C}}$ the Stein exponent is zero for all $\epsilon \in [0, 1)$.

Theorems 2 and 4 also apply to the k-fold product channel $\mathcal{N}^{\otimes k}$. If $\mathcal{N}^{\otimes k} \in \bar{\mathcal{C}}$, then $\theta_k^\star(\epsilon) = 0$ for all $\epsilon \in [0, 1)$. If $\mathcal{N}^{\otimes k} \in \mathcal{C}$, then $\theta_k^\star(\epsilon) = D(\rho\|\sigma)$ for all ϵ above a channel-dependent threshold; the exponent below that threshold is not determined by these results. For large values of k, the condition whether $\mathcal{N}^{\otimes k} \in \mathcal{C}$ or $\mathcal{N}^{\otimes k} \in \bar{\mathcal{C}}$ can however be cumbersome to check, and we seek for conditions that can directly be checked on the elementary channel $\mathcal{N}$.

The rest of this section focuses on the question whether for a channel $\mathcal{N} \in \bar{\mathcal{C}}$ the two-fold product $\mathcal{N}^{\otimes 2}$ lies in $\mathcal{C}$ or in $\bar{\mathcal{C}}$. Inspired by Lemma 2 and Remark 1, see (23), we make the following definition.

Definition 3. *We say that a quantum channel $\mathcal{N}$ can* one-sided zero-error self-superactivate *if $\mathcal{N} \in \bar{\mathcal{C}}$ but $\mathcal{N}^{\otimes 2} \in \mathcal{C}$.*

Above definition is similar to the definition of *zero-error capacity superactivation* reported in [4], where two channels superactivate if for both channels the orthogonal complement of the space $\text{Span}\{\mathsf{K}_i^\dagger \mathsf{K}_j\}_{i,j}$ does not contain a rank-one matrix (and thus both channels have vanishing one-shot zero-error capacity), while for the tensor product of the two channels the corresponding orthogonal space contains a rank-one matrix.

Our last two results are a negative and a positive result on one-sided zero-error self-superactivation.

Theorem 5. *Any quantum channel $\mathcal{N}\colon \mathbb{C}^{N_{\text{in}} \times N_{\text{in}}} \to \mathbb{C}^{N_{\text{out}} \times N_{\text{out}}}$ in $\bar{\mathcal{C}}$ with*

$$\dim(\mathcal{K}_{\mathcal{N}}) = N_{\text{in}} N_{\text{out}} - 1, \tag{30}$$

cannot one-sided zero-error self-superactivate, i.e., also the two-fold channel $\mathcal{N}^{\otimes 2}$ lies in $\bar{\mathcal{C}}$. Thus

$$\theta_2^\star(\epsilon) = 0. \tag{31}$$

Proof. See Section IV-E. □

Corollary 6. *Any qubit-channel $\mathcal{N}\colon \mathbb{C}^{2\times 2} \to \mathbb{C}^{2\times 2}$ cannot one-sided zero-error self-superactivate.*

Proof. Qubit channels have $\dim(\mathcal{K}_{\mathcal{N}}) \in \{1, 2, 3, 4\}$. For channels with $\dim(\mathcal{K}_{\mathcal{N}}) = 4$, $\mathcal{N}$ lies in $\mathcal{S}^\circ$ and the desired result follows from Theorem 1. For channels with $\dim(\mathcal{K}_{\mathcal{N}}) \leq 2$, the desired result holds by Corollary 3. Finally, for channels with $\dim(\mathcal{K}_{\mathcal{N}}) = 3$, the result follows from above Theorem 5. □

In contrast, the channel $\mathcal{N}_{\text{W}}\colon \mathbb{C}^{4\times 4} \to \mathbb{C}^{4\times 4}$ described by the eight Kraus operators

$$(\mathsf{K}_1, \ldots, \mathsf{K}_8) = \frac{1}{\sqrt{8}}\big(\mathsf{I}_4, \mathsf{Q}^2, \mathsf{Q}^3, \mathsf{P}^3, \mathsf{P}^2\mathsf{Q}, \mathsf{P}^3\mathsf{Q}^3, \mathsf{P}\mathsf{Q}, \mathsf{P}^3\mathsf{Q}^2\big), \tag{32}$$

can one-sided zero-error self-superactivate, where in the above

$$\mathsf{P} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad \mathsf{Q} = \text{diag}(1, i, -1, -i). \tag{33}$$

Since each of the Kraus operators is proportional to a unitary matrix, $\sum_{j=1}^8 \mathsf{K}_j^\dagger \mathsf{K}_j = \sum_{j=1}^8 \mathsf{K}_j \mathsf{K}_j^\dagger = \mathsf{I}_4$, and $\mathcal{N}_{\text{W}} \in \mathcal{S}$.

Theorem 7. *The channel $\mathcal{N}_{\text{W}}$ can one-sided zero-error self-superactivate. I.e.,*

$$\mathcal{N}_{\text{W}} \in \bar{\mathcal{C}}, \qquad \mathcal{N}_{\text{W}}^{\otimes 2} \in \mathcal{C}. \tag{34}$$

Proof. See Section IV-F. □

IV. Proofs

A. Proof of Lemma 1

Fix arbitrary encoding and decision sequences $(\{\mathsf{L}_i\}_i, \mathsf{M}_0, \mathsf{M}_1)$ such that

$$\overline{\lim_{n\to\infty}}\ \alpha_n \leq \epsilon. \tag{35}$$

We first prove that (35) implies

$$\overline{\lim_{n\to\infty}}\ \text{Tr}\,(\mathsf{M}_0) \geq 1 - \epsilon. \tag{36}$$

This can be seen by noting that

$$1 - \alpha_n = \text{Tr}\,(\mathsf{M}_0 \mathsf{A}_{\text{out}}^\rho) \leq \text{Tr}\,(\mathsf{M}_0), \tag{37}$$

where the last inequality holds because the eigenvalues of $\mathsf{A}_{\text{out}}^{\rho}$ are upper bounded by 1. In fact, by the Ruhe-Von Neumann trace inequality [11], we have

$$\operatorname{Tr}\left(\mathsf{M}_0\mathsf{A}_{\text{out}}^{\rho}\right) \leq \sum_{i=1}^{N_{\text{out}}^k} c_i d_{N_{\text{out}}^k-i+1} \leq \sum_{i=1}^{N_{\text{out}}^k} c_i = \operatorname{Tr}\left(\mathsf{M}_0\right), \tag{38}$$

for c_i and d_i the eigenvalues of the positive semidefinite matrices M_0 and $\mathsf{A}_{\text{out}}^{\rho}$ in ascending order, respectively.

Now, notice that by Von Neumann's trace inequality [14], and by denoting the eigenvalues of $\mathsf{A}_{\text{out}}^{\sigma}$ by e_i in ascending order, we obtain:

$$\begin{aligned}
\beta_n &= \operatorname{Tr}\left(\mathsf{M}_0\mathsf{A}_{\text{out}}^{\sigma}\right) && (39)\\
&\geq \sum_{i=1}^{N_{\text{out}}^k} c_i e_{N_{\text{out}}^k-i+1} && (40)\\
&\geq e_1 \sum_{i=1}^{N_{\text{out}}^k} c_i && (41)\\
&\geq \gamma_{\mathcal{N},k} \cdot \operatorname{Tr}\left(\mathsf{M}_0\right), && (42)
\end{aligned}$$

with

$$\gamma_{\mathcal{N},k} := \inf_{\substack{\tau\succeq 0:\ \tau\neq 0\\ \operatorname{Tr}(\tau)=1}} c_{\min}(\mathcal{N}^{\otimes k}(\tau)). \tag{43}$$

Notice the set of input density matrices is compact. Moreover, $\mathcal{N}$ is continuous, and the smallest eigenvalue of a Hermitian matrix is a continuous function of that matrix. The infimum in (43) is therefore attained at some density matrix τ^*, and thus indeed a minimum.

This concludes the proof of the lemma.

B. Proof of Theorem 1

We first prove the statement for the *completely depolarizing channel*

$$\mathcal{N}_{\mathcal{D}}\colon \tau \mapsto \frac{\mathsf{I}_{N_{\text{out}}}}{N_{\text{out}}} \cdot \operatorname{Tr}\left(\tau\right), \tag{44}$$

and then extend the argument to any quantum channel $\mathcal{N} \in \mathcal{Q}^{\circ}$.

The k-fold product of the depolarizing channel is the mapping:

$$\mathcal{N}_{\mathcal{D}}^{\otimes k}\colon \tau_k \mapsto \frac{\mathsf{I}_{N_{\text{out}}^k}}{N_{\text{out}}^k} \cdot \operatorname{Tr}\left(\tau_k\right), \tag{45}$$

and thus

$$\gamma_{\mathcal{N}_{\mathcal{D}},k} = \frac{1}{N_{\text{out}}^k}, \tag{46}$$

which is strictly positive and moreover satisfies

$$\lim_{n\to\infty} \frac{1}{n} \log \gamma_{\mathcal{N}_{\mathcal{D}},k(n)} = 0 \tag{47}$$

whenever $\frac{k(n)}{n} \to 0$. Combining (47) with Lemma 1 establishes the desired result $\theta^*_{\text{sublin}}(\epsilon) = 0$ for the completely depolarizing channel.

Consider now an arbitrary quantum channel $\mathcal{N} \in \mathcal{Q}^{\circ}$. Since $\mathcal{Q}$ is convex, $\mathcal{N}$ can be expressed as a convex combination of $\mathcal{N}_{\mathcal{D}}$ and some other channel $\mathcal{N}' \in \mathcal{Q}$:

$$\mathcal{N} = \nu\mathcal{N}_{\mathcal{D}} + (1-\nu)\mathcal{N}' \tag{48}$$

for $\nu \in [0,1]$ and $\nu \neq 0$ when $\mathcal{N}$ is not the completely depolarizing channel.

Notice that for any input state τ_k of the k-fold product channel $\mathcal{N}^{\otimes k}$, we obtain as output state:

$$\begin{aligned}
\mathcal{N}^{\otimes k}(\tau_k) &= \sum_{\mathcal{T}\in\{\mathcal{N}_{\mathcal{D}},\mathcal{N}'\}^{\otimes k}} \nu^{d_{\mathcal{T}}}(1-\nu)^{n_{\mathcal{T}}}\mathcal{T}(\tau_k) && (49)\\
&= \nu^k \mathcal{N}_{\mathcal{D}}^{\otimes k}(\tau_k) && (50)\\
&\quad + \underbrace{\sum_{\mathcal{T}\in\{\mathcal{N}_{\mathcal{D}},\mathcal{N}'\}^{\otimes k}\setminus\mathcal{N}_{\mathcal{D}}^{\otimes k}} \nu^{d_{\mathcal{T}}}(1-\nu)^{n_{\mathcal{T}}}\mathcal{T}(\tau_k)}_{\succeq 0} && (51)\\
&\succeq \nu^k \frac{\mathsf{I}_{N_{\text{out}}^k}}{N_{\text{out}}^k} && (52)
\end{aligned}$$

where $d_{\mathcal{T}}$ and $n_{\mathcal{T}}$ denote the numbers of occurrences of the channels $\mathcal{N}_{\mathcal{D}}$ and $\mathcal{N}'$ in the product channel $\mathcal{T}$, respectively.

We conclude that

$$\gamma_{\mathcal{N},k} \geq \left(\frac{\nu}{N_{\text{out}}}\right)^k, \tag{53}$$

and thus again by Lemma 1 we can deduce that $\theta^*_{\text{sublin}} = 0$ for all channels in the relative interior $\mathcal{Q}^{\circ}$ of $\mathcal{Q}$.

C. Proof of Theorem 2

In view of (24), we only need to show achievability:

$$\theta_k^*(\epsilon), \theta^{\star}_{\text{sublin}}(\epsilon) \geq D(\rho\|\sigma). \tag{54}$$

By the Quantum Stein Lemma [9], there exists a sequence of measurements $(\Pi_{0,n}, \Pi_{1,n})$ such that

$$\alpha_{n,\text{local}} := \operatorname{Tr}\left(\Pi_{1,n}\rho^{\otimes n}\right) \tag{55}$$

$$\beta_{n,\text{local}} := \operatorname{Tr}\left(\Pi_{0,n}\sigma^{\otimes n}\right) \tag{56}$$

satisfy

$$\overline{\lim_{n\to\infty}}\ \alpha_{n,\text{local}} = 0 \tag{57}$$

$$-\overline{\lim_{n\to\infty}}\ \frac{1}{n}\log\beta_{n,\text{local}} = D(\rho\|\sigma). \tag{58}$$

Choose two pure input states ϕ and ψ and a measurement $\{\tilde{\mathsf{M}}_0, \tilde{\mathsf{M}}_1 = \mathsf{I} - \tilde{\mathsf{M}}_0\}$ so that (existence was shown in (23))

$$\operatorname{Tr}\left(\tilde{\mathsf{M}}_0\mathcal{N}(\psi)\right) = 0 \tag{59}$$

$$\operatorname{Tr}\left(\tilde{\mathsf{M}}_0\mathcal{N}(\phi)\right) > 0. \tag{60}$$

Fix a blocklength n. For this blocklength, apply the measurement $(\Pi_{0,n}, \Pi_{1,n})$ and denote the measurement outcome B. For outcome $B = 0$ send

$$\mathsf{A}_{\text{in}} = \phi^{\otimes k} \tag{61}$$

while for outcome $B = 1$ send

$$\mathsf{A}_{\text{in}} = \psi^{\otimes k}. \tag{62}$$

With this choice, the channel input states under the two hypotheses can be written as:

$$\mathsf{A}_{\text{in}}^{\rho} = (1-\alpha_{n,\text{local}})\cdot\phi^{\otimes k} + \alpha_{n,\text{local}}\cdot\psi^{\otimes k} \tag{63}$$

$$\mathsf{A}_{\text{in}}^{\sigma} = \beta_{n,\text{local}} \cdot \phi^{\otimes k} + (1-\beta_{n,\text{local}}) \cdot \psi^{\otimes k} \tag{64}$$

The decision center applies measurement $(\mathsf{I} - \tilde{\mathsf{M}}_1^{\otimes k}, \tilde{\mathsf{M}}_1^{\otimes k})$ to the output state A_{out}. The outcome of this binary measurement is $\hat{\mathcal{H}}$.

We analyze the error probabilities of this scheme. For the type-II error probability we have:

$$\begin{aligned}
\beta_n &= 1 - \text{Tr}\left(\tilde{\mathsf{M}}_1^{\otimes k} \mathcal{N}^{\otimes k}\left(\mathsf{A}_{\text{in}}^{\sigma}\right)\right) && (65)\\
&= 1 - \beta_{n,\text{local}} \cdot \text{Tr}\left(\tilde{\mathsf{M}}_1^{\otimes k} \mathcal{N}^{\otimes k}\left(\phi^{\otimes k}\right)\right) \\
&\quad -(1-\beta_{n,\text{local}}) \cdot \text{Tr}\left(\tilde{\mathsf{M}}_1^{\otimes k} \mathcal{N}^{\otimes k}\left(\psi^{\otimes k}\right)\right) && (66)\\
&= \quad 1 - \beta_{n,\text{local}} \cdot \Big(\underbrace{\text{Tr}\left(\tilde{\mathsf{M}}_1 \mathcal{N}(\phi)\right)}_{<1}\Big)^k \\
&\quad -(1-\beta_{n,\text{local}}) \cdot \Big(\underbrace{\text{Tr}\left(\tilde{\mathsf{M}}_1 \mathcal{N}(\psi)\right)}_{=1}\Big)^k && (67)\\
&\leq \beta_{n,\text{local}}. && (68)
\end{aligned}$$

and

$$-\varlimsup_{n\to\infty} \frac{1}{n} \log \beta_n \geq -\varlimsup_{n\to\infty} \frac{1}{n} \log \beta_{n,\text{local}} = D(\rho\|\sigma), \tag{69}$$

where the last equality holds by (58).

Similarly, for the type-I error probability:

$$\begin{aligned}
\alpha_n &= \text{Tr}\left(\tilde{\mathsf{M}}_1^{\otimes k} \mathcal{N}^{\otimes k}(\mathsf{A}_{\text{in}}^{\rho})\right) && (70)\\
&= (1-\alpha_{n,\text{local}}) \cdot \text{Tr}\left(\tilde{\mathsf{M}}_1^{\otimes k} \mathcal{N}^{\otimes k}\left(\phi^{\otimes k}\right)\right) \\
&\quad +\alpha_{n,\text{local}} \cdot \text{Tr}\left(\tilde{\mathsf{M}}_1^{\otimes k} \mathcal{N}^{\otimes k}\left(\psi^{\otimes k}\right)\right) && (71)\\
&= (1-\alpha_{n,\text{local}}) \cdot \Big(\underbrace{\text{Tr}\left(\tilde{\mathsf{M}}_1 \mathcal{N}(\phi)\right)}_{<1}\Big)^k \\
&\quad +\alpha_{n,\text{local}} \cdot \Big(\underbrace{\text{Tr}\left(\tilde{\mathsf{M}}_1 \mathcal{N}(\psi)\right)}_{=1}\Big)^k && (72)\\
&= (1-\alpha_{n,\text{local}}) \cdot \text{Tr}\left(\tilde{\mathsf{M}}_1 \mathcal{N}(\phi)\right)^k + \alpha_{n,\text{local}}. && (73)
\end{aligned}$$

Defining

$$q := \text{Tr}\left(\tilde{\mathsf{M}}_1 \mathcal{N}(\phi)\right) \in [0,1), \tag{74}$$

where $q < 1$ by (60), we continue on (73) to conclude

$$\alpha_n = (1-\alpha_{n,\text{local}}) q^k + \alpha_{n,\text{local}}. \tag{75}$$

For fixed k, (75) and (57) imply

$$\lim_{n\to\infty} \alpha_n = q^k. \tag{76}$$

Combining (69) and (76) proves the theorem for every ϵ satisfying $q^k \leq \epsilon < 1$.

If $k = k(n) \to \infty$, the desired result follows as above by noting that $q^{k(n)} \to 0$ as $n \to \infty$.

D. Proof of Theorem 4,

By definition, $\theta_1^*(\epsilon) \geq 0$, and we only need to prove $\theta_1^*(\epsilon) \leq 0$.

To prove this latter statement, notice that for $\mathcal{N} \in \bar{\mathcal{C}}$:

$$\gamma_{\mathcal{N},1} > 0, \tag{77}$$

independent of the blocklength n, because for channels $\mathcal{N} \in \bar{\mathcal{C}}$ every input density matrix induces a full-rank output state with all strictly positive eigenvalues. In particular,

$$\lim_{n\to\infty} \frac{1}{n} \log \gamma_{\mathcal{N},1} = 0, \tag{78}$$

and thus the desired statement follows immediately from Lemma 1.

E. Proof of Theorem 5

By assumption the orthogonal Kraus space $\mathcal{K}^{\perp}$ is of dimension 1, and thus

$$\mathcal{K}^{\perp} = \text{Span}\{\mathsf{A}\}, \tag{79}$$

for some matrix A. Since the channel $\mathcal{N} \in \bar{\mathcal{C}}$, we have

$$r := \text{rank}(\mathsf{A}) \geq 2 \tag{80}$$

Next observe that the orthogonal complement of the Kraus space $\mathcal{K}_{\mathcal{N}^{\otimes 2}}$ is

$$\mathcal{K}_{\mathcal{N}^{\otimes 2}}^{\perp} = \text{Span}\{\mathsf{A}\} \otimes \mathbb{C}^{N_{\text{out}} \times N_{\text{in}}} + \mathbb{C}^{N_{\text{out}} \times N_{\text{in}}} \otimes \text{Span}\{\mathsf{A}\}. \tag{81}$$

Hence, every element of this space has the form

$$\mathsf{Z} = \mathsf{A} \otimes \mathsf{X} + \mathsf{Y} \otimes \mathsf{A}, \qquad \mathsf{X}, \mathsf{Y} \in \mathbb{C}^{N_{\text{out}} \times N_{\text{in}}}. \tag{82}$$

Choose now invertible matrices L and R of appropriate sizes such that

$$\mathsf{LAR} = \mathsf{D}_r, \tag{83}$$

where $\mathsf{D}_r \in \mathbb{C}^{N_{\text{out}} \times N_{\text{in}}}$ consists of the identity matrix I_r on its upper-left corner and otherwise has all zero entries. Since the invertible transformation

$$\mathsf{Z} \mapsto (\mathsf{L} \otimes \mathsf{L}) \mathsf{Z} (\mathsf{R} \otimes \mathsf{R}) \tag{84}$$

preserves rank, in the following we only consider Z-matrices of the form

$$\mathsf{Z} = \mathsf{D}_r \otimes \mathsf{X} + \mathsf{Y} \otimes \mathsf{D}_r,$$

and show that it cannot be of rank 1.

Regard Z as an $N_{\text{out}} \times N_{\text{in}}$ block matrix with blocks Z_{ij} of dimension $N_{\text{out}} \times N_{\text{in}}$. Denoting the elements of Y by y_{ij}, we can write

$$\mathsf{Z}_{ij} = (\mathsf{D}_r)_{ij} \mathsf{X} + y_{ij} \mathsf{D}_r.$$

For the sake of contradiction, assume that $\text{rank}(\mathsf{Z}) \leq 1$. Every block then has rank at most one. Whenever $(\mathsf{D}_r)_{ij} = 0$, we have $\mathsf{Z}_{ij} = y_{ij}\mathsf{D}_r$; since $\text{rank}(\mathsf{D}_r) = r \geq 2$, this forces $y_{ij} = 0$. Consequently, the only entries of Y that can be nonzero are $y_{11}, \ldots, y_{rr}$, and the only potentially nonzero blocks of Z are

$$\mathsf{Z}_{ii} = \mathsf{X} + y_{ii}\mathsf{D}_r, \qquad 1 \leq i \leq r.$$

They occupy mutually disjoint block rows and block columns.

Since $r \geq 2$ and $\mathrm{rank}(\mathsf{Z}) \leq 1$, at most one of these blocks can be nonzero. Thus at least one, say the j-th, vanishes:

$$\mathsf{X} + y_{jj}\mathsf{D}_r = 0.$$

Hence $\mathsf{X} = -y_{jj}\mathsf{D}_r$, and every remaining block is

$$\mathsf{Z}_{ii} = (y_{ii} - y_{jj})\mathsf{D}_r.$$

Each such block has rank either $r = 0$ or $r \geq 2$. The assumption $\mathrm{rank}\,(Z) \leq 1$ therefore forces every block to vanish. Thus $\mathsf{Z} = 0$ and the space $\mathcal{K}_{\mathcal{N}^{\otimes 2}}^{\perp}$ contains no nonzero rank-one matrix, which concludes the proof.

F. Proof of Theorem 7

Let P, Q, and $\mathsf{K}_1, \ldots, \mathsf{K}_8$ be as defined before Theorem 7. The Weyl matrices $\mathsf{W}_{a,b} = \mathsf{P}^a\mathsf{Q}^b$, $a, b \in \{0, 1, 2, 3\}$, form an orthogonal basis of $\mathbb{C}^{4\times 4}$ for the Hilbert–Schmidt inner product. Since $\mathcal{K}_{\mathcal{N}_\mathrm{W}} = \mathrm{span}\{\mathsf{K}_1, \ldots, \mathsf{K}_8\}$, orthogonality gives

$$\mathcal{K}_{\mathcal{N}_\mathrm{W}}^{\perp} = \mathrm{span}\{\mathsf{Q}, \mathsf{P}, \mathsf{P}\mathsf{Q}^2, \mathsf{P}\mathsf{Q}^3, \mathsf{P}^2, \mathsf{P}^2\mathsf{Q}^2, \mathsf{P}^2\mathsf{Q}^3, \mathsf{P}^3\mathsf{Q}\}.$$

Writing an arbitrary $\mathsf{Z} \in \mathcal{K}_{\mathcal{N}_\mathrm{W}}^{\perp}$ in this basis and expanding gives

$$\mathsf{Z} = \begin{pmatrix} c_0 & ic_7 & c_4 + c_5 - c_6 & c_1 - c_2 + ic_3 \\ c_1 + c_2 + c_3 & ic_0 & -c_7 & c_4 - c_5 + ic_6 \\ c_4 + c_5 + c_6 & c_1 - c_2 - ic_3 & -c_0 & -ic_7 \\ c_7 & c_4 - c_5 - ic_6 & c_1 + c_2 - c_3 & -ic_0 \end{pmatrix}.$$

If $\mathrm{rank}(\mathsf{Z}) \leq 1$, all 2×2 minors vanish. To verify that this forces $\mathsf{Z} = 0$, let Δ_{rs}^{jk} denote the 2×2 minor in rows r, s and columns j, k of Z. Direct expansion gives

$$c_0^2 = \frac{i}{4}(-\Delta_{12}^{12} + \Delta_{14}^{14} + \Delta_{23}^{23} - \Delta_{34}^{34}), \tag{85}$$

$$c_3^2 = \frac{i}{4}(\Delta_{12}^{14} - \Delta_{14}^{34} + \Delta_{23}^{12} - \Delta_{34}^{23}), \tag{86}$$

$$c_6^2 = \frac{i}{4}(\Delta_{12}^{34} + \Delta_{14}^{23} + \Delta_{23}^{14} + \Delta_{34}^{12}), \tag{87}$$

$$c_7^2 = \frac{i}{4}(\Delta_{12}^{23} + \Delta_{14}^{12} - \Delta_{23}^{34} - \Delta_{34}^{14}). \tag{88}$$

Consequently, all these coefficients vanish. The remaining matrix is

$$\mathsf{Z} = \begin{pmatrix} 0 & 0 & c_4 + c_5 & c_1 - c_2 \\ c_1 + c_2 & 0 & 0 & c_4 - c_5 \\ c_4 + c_5 & c_1 - c_2 & 0 & 0 \\ 0 & c_4 - c_5 & c_1 + c_2 & 0 \end{pmatrix}. \tag{89}$$

Its minors satisfy

$$\Delta_{13}^{13} = -(c_4 + c_5)^2, \qquad \Delta_{24}^{24} = -(c_4 - c_5)^2, \tag{90}$$

$$\Delta_{13}^{24} = -(c_1 - c_2)^2, \qquad \Delta_{24}^{13} = (c_1 + c_2)^2. \tag{91}$$

Their vanishing implies $c_1 = c_2 = c_4 = c_5 = 0$, and thus $\mathsf{Z} = 0$. Thus $\mathcal{K}_{\mathcal{N}_\mathrm{W}}^{\perp}$ contains no nonzero rank-one matrix and $\mathcal{N}_\mathrm{W} \in \bar{\mathcal{C}}$.

Let now

$$\mathsf{A} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \tag{92}$$

and notice that

$$\mathrm{Tr}\left(\mathsf{K}_j\mathsf{A}\mathsf{K}_k^T\right) = 0, \qquad 1 \leq j, k \leq 8. \tag{93}$$

Define then the vectors

$$|\Phi^+\rangle = \mathrm{vec}(\mathsf{I}_4) \qquad |\psi\rangle = \mathrm{vec}(\mathsf{A}), \tag{94}$$

and the derived nonzero rank-one matrix

$$\mathsf{F} = |\Phi^+\rangle\langle\psi|. \tag{95}$$

Since for every $j, k \in \{1, \ldots, 8\}$:

$$\mathrm{Tr}\left(\mathsf{F}^\dagger(\mathsf{K}_j \otimes \mathsf{K}_k)\right) = \mathrm{Tr}\left(\mathsf{K}_j\mathsf{A}\mathsf{K}_k^T\right) = 0, \tag{96}$$

where we used $\mathsf{A}^\dagger = \mathsf{A}$ and (93), and since the matrices $\mathsf{K}_j \otimes \mathsf{K}_k$ span $\mathcal{K}_{\mathcal{N}_\mathrm{W}} \otimes \mathcal{K}_{\mathcal{N}_\mathrm{W}}$, it follows that

$$\mathsf{F} \in (\mathcal{K}_{\mathcal{N}_\mathrm{W}} \otimes \mathcal{K}_{\mathcal{N}_\mathrm{W}})^{\perp}, \tag{97}$$

which proves that

$$\mathcal{N}_\mathrm{W}^{\otimes 2} \in \mathcal{C}. \tag{98}$$

CONCLUSION

We proved a dichotomy of the Stein exponent in a remote quantum hypothesis testing problem where the sensor communicates over a quantum channel to the decision center. Depending on whether the quantum channel's orthogonal Kraus space contains a rank-one element or not, the Stein exponent equals the one over a perfect channel (for sufficiently large type-I error thresholds) or degrades to zero (for all type-I error thresholds). We also discuss a self-superactivation phenomenon and present a channel with four input and output dimensions where the Stein exponent with a single channel use is zero, but with two channel uses the perfect-channel exponent is achievable. In contrast, we show that this superactivation phenomenon cannot happen for certain classes of channels, including qubit channels.

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